

Competing Explanations, Persisting Rules: Grounded Defense against Misleading Charts in Executing Web Agents

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Paper ID: NAACL-2027-XXXX

Abstract

Vision-language agents that execute multi-step web tasks treat charts as evidence: the way a chart is read enters tool calls, submissions, and actions. Charts, however, admit readings both faithful to the rendered image and wrong about the underlying data, and existing defenses operate on single question-answer rounds, not how a revised interpretation remains in force over stateful execution. We present CERS (Competing-Explanation Rule State), a harness-level defense that organizes chart interpretation around two coupled mechanisms: a competing-explanation chain that expresses the current judgment as $O \wedge B \Rightarrow C_q(a)$ —an observation, an interpretation rule, and the conclusion an action requires—and a persistent rule state that records every validated rule revision with its evidence and scope, re-using it in later reasoning. On a paired static panel of 140 executable web tasks built on MISVISAGENTBENCH, CERS raises the robust accuracy of the Qwen3.8-27B open-weight VLM agent from 57.9% to 68.6%, replicates the gain in a frozen rerun (52.1% to 64.3%; McNemar $p = .0013$), holds it across request seeds. It holds against the baselines ($65.7\% \pm 0.7$ against $56.2\% \pm 1.1$; pooled $p = 1.9 \times 10^{-5}$), and transfers with a smaller gain to an 8B agent (46.4% to 53.6%). In stateful online execution over twelve live tasks spanning both chart variants, the ordinary agent submits six wrong actions, all on misleading charts under the 27B actor, while the CERS arm submits none in any batch or chart variant; in the frontier-model batches the defense arm converts correct misleading-chart decisions into recorded abstentions (exact sign test $p = .002$) rather than wrong submissions, and a stability rerun keeps 37 of 48 unit outcomes on the same side of the submission boundary.

1 Introduction

Large language model agents act by looping through observation, reasoning, and action in an external environment; what the agent believes about

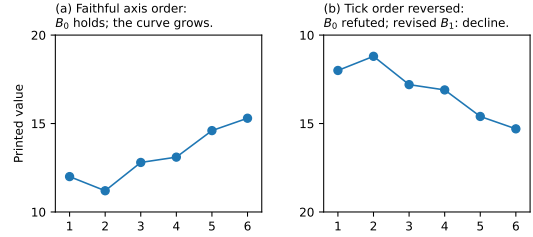


Figure 1: The same drawn series under two printed axis orders: the customary rule B_0 holds in (a) and is refuted by the scale order in (b). Model inputs never reveal which panel is which.

the information it reads enters tool calls and operations directly, making reliable interpretation of external content a precondition for execution safety. Prior work shows that indirect prompt injections in web pages and tool returns steer agents away from the user’s intent (Zhan et al., 2024; Evtimov et al., 2025; Johnson et al., 2025), that visual perturbations hijack multi-step web agents (Wu et al., 2024), and that deceptive interface design can be probed systematically (Wang et al., 2025): an environment intervenes not only through explicit instructions but through the way the agent understands information, shaping the decisions it makes autonomously.

Charts are prominent among such inputs, and correct and misleading readings can coexist in the same figure: charts support misleading inferences even when drawn faithfully (Lisnic et al., 2023; Nadib et al., 2026). A curve moving upward is observed accurately, but whether it represents growth depends on the axis mapping in force; a misleading presentation can thus induce an agent to apply its customary reading rules unchanged and convert visible evidence into a wrong conclusion (Figure 1).

Multimodal models are systematically vulnerable to such manipulations in chart question answering (Chen et al., 2025; Mahbub et al., 2025, 2026), and our companion work under review formalizes

the execution-level consequence: biases induced by misleading charts propagate into subsequent actions and change task outcomes on executable web tasks; direct selection recovers 26 accuracy points on the paired faithful charts (rate gap on seed means; paired net +33; Section 4.2).¹

Against misleading charts, model-level defenses have improved chart question answering through tabular answering and re-plotting (Tonglet et al., 2026), but the reliability of verification itself remains open: intrinsic self-correction can damage initially correct answers (Huang et al., 2023; Li, 2025), reflection and search open new attack surfaces when the evaluator is exposed to the environment (Wu et al., 2024), and a model may identify chart properties correctly yet fail to let them govern downstream inference (Mahbub et al., 2026). Two requirements follow: correction must rest on checkable evidence, and a corrected interpretation must actually change subsequent judgments—an added review round can re-adopt an inapplicable reading as easily as discover one.

Stateful execution raises the stakes: an understanding formed from one chart feeds later planning even after the chart has left the viewport, so a verified mapping should keep governing later judgments while staying revisable on new evidence. Reflexion-style memory carries verbalized feedback into later attempts (Shinn et al., 2023), and agent-memory systems organize experience into retrievable state (Wang et al., 2024; Hu et al., 2025); an inductive chart requires something more specific, the current validity of interpretations: which rules have evidential support, which have been refuted, and what new evidence suffices to change a reading? We pose one unified question: *how can an agent reliably revise visual interpretation rules, and keep validated rules in force over subsequent execution?*

We answer this question with a defense that combines *competing explanation chains* with a *persistent rule state*, harness-level externalization (Zhou et al., 2026a) put to defensive use. When the agent forms a chart-dependent judgment, the harness expresses it as $O \wedge B \Rightarrow C_q(a)$: an observation O in the chart, an interpretation rule B that turns the observation into task meaning, and the conclusion $C_q(a)$ that an action a requires under the task q . The

harness then builds candidate chains with different interpretive bases on the same complete chart, turning default visual conventions into explicit objects of test, and compares their applicability using evidence drawn from the figure itself. Validated rule revisions enter the agent’s runtime harness as an interpretation state that later reasoning reads and updates by scope; later decisions reuse the rules in force, and re-enter verification only when new effective evidence challenges them. Competing explanations determine why an interpretation needs to change; the persistent state carries that evidenced change into later actions.

Contributions: (1) a correction framework based on competing explanation chains, where $O \wedge B \Rightarrow C_q(a)$ unifies visual observation, interpretation rules, and action judgments, and customary readings are tested against in-chart evidence, so the rules behind wrong inferences are identified and revised. (2) A stateful defense for continuing execution maintains validated rules, revision evidence, and scope conditions as task-scoped harness state, so later decisions reuse confirmed readings and revise them only on new effective evidence.

2 Related Work

Research on misleading visualizations connects two lines. The first studies how charts mislead human readers: deception relies less on outright distortion than on faithful drawings with selective context (Lisnic et al., 2023; Nadib et al., 2026). The second measures the phenomenon in models: benchmarks evaluate multimodal LLMs on misleading chart QA (Chen et al., 2025; Bharti et al., 2024; Alexander et al., 2024), bias across interpretations is documented (Rahman et al., 2026), and agent recognition of misleadingness is probed (Panda, 2026; Blasilli and Angelini, 2026; Seto et al., 2026); manipulations are quantified with mitigations (Mahbub et al., 2025, 2026), and method-side responses include tool-augmented reasoning (Kaur et al., 2026), structured extraction (Das et al., 2025), detection (Tonglet et al., 2025), and dual-path frameworks (Zhang et al., 2026). The closest model-level defense, tabular answering and re-plotting (Tonglet et al., 2026), repairs a single question–answer round, while CERS targets the interpretation rules that recur across an executing agent’s rounds and maintains their validated revisions in runtime state.

A second line concerns agent-environment in-

¹Companion submission under review; the paired benchmark MISVISAGENTBENCH and the execution-level evaluation protocol it introduces are described where first used in this paper.

tegrity: injection benchmarks show that tool and web content can steer agents (Zhan et al., 2024), security benchmarks measure targeted hijacking (Evtimov et al., 2025), injection reaches perceived representations such as HTML accessibility trees (Johnson et al., 2025), red-teaming and runtime frameworks generate and intercept attacks at scale (Wang et al., 2025; Zhao et al., 2026), and interface-level defenses separate instructions from data (Chen et al., 2024a) or train against injected content (Chen et al., 2024b); reflection and tree search open new vulnerabilities when the evaluator is exposed to attacked content (Wu et al., 2024). CERS protects a different layer: not the instruction hierarchy but the interpretation layer between visual observation and task judgment, where a misleading chart exerts its effect without any injected instruction.

The third line concerns persistent state and correction: verbal reinforcement carries feedback across attempts (Shinn et al., 2023), self-consistency aggregates reasoning paths (Wang et al., 2022), intrinsic self-correction without external feedback degrades reasoning (Huang et al., 2023; Li, 2025), surveys organize the state and memory design space (Wang et al., 2024; Hu et al., 2025; Zhou et al., 2026b), and realistic visual web tasks remain hard enough that single-decision correctness cannot be taken for granted (Koh et al., 2024). CERS borrows the state-carrying idea but changes the carried content: the state records interpretation rules with support, scope, and revision history, so later steps inherit what was verified about the chart, not a verdict about the answer.

3 Method

CERS rests on one object: the interpretation rule B that connects visual observation to task judgment. Competing explanation chains make these rules explicit and test them; persistent state organizes their revision evidence, current validity, and applicability scope, so that they participate in later interaction. This section develops the two mechanisms in turn after fixing the task setting.

3.1 Task Modeling and Overall Framework

Consider an agent executing a task q : at step t it receives an observation o_t and, given history H_t , proposes $\tilde{a}_t$. When the proposal depends on chart understanding, let I_t denote the complete chart observation relevant to the judgment—or the chart observation obtained earlier in the task when the

chart has left the current page. We express a chart-driven interpretation as

$$O \wedge B \Rightarrow C_q(a), \quad (1)$$

where O is an observation in the chart, B is the interpretation rule required to convert the observation into task semantics, and $C_q(a)$ is the proposition that the action, or high-level choice, a meets the requirements of task q . The representation separates a visible phenomenon from the condition under which it is interpretable: a curve moving upward, jointly with the rule that higher position means larger value in the current axis, supports the conclusion that the interval grows and the action is admissible, while a mismatch against verified numeric relations refutes precisely that rule’s applicability. O and B may carry multiple items covering entity binding, numeric relations, encoding direction, time range, and comparison conditions; explicitly computable relations are checked with program assistance, and remaining semantic judgments are verified jointly with the public task.

The method maintains a within-task rule state M_t^B and, for chart-dependent judgments, runs the process *current interpretation* $\rightarrow$ *competing-explanation generation* $\rightarrow$ *in-chart applicability verification* $\rightarrow$ *rule-state update* $\rightarrow$ *action re-derivation*; later steps read the applicable rules from the current context and restart competing verification only when a relevant judgment is first formed, when new evidence conflicts with an existing rule, or when a rule’s applicability conditions are in question. Local verification decides acceptance or revision; within-task maintenance decides where a rule remains effective. The model’s parameters stay frozen; the rule state evolves only with observations and verification results actually obtained.

3.2 Competing Explanation Chains

3.2.1 Making the Current Explanation Explicit

When the agent proposes a chart-dependent choice $\tilde{a}_t$, the interpretation module forms, from the complete chart, the public task, the admissible history, and the current rule state,

$$\pi_0 = \langle O_0, B_0, p_0 \rangle, \quad p_0 = C_q(\tilde{a}_t), \quad (2)$$

stating which phenomena the chart is claimed to contain, which rules carry the observation to the conclusion, and whether the required evidence is

present; unsupported choices record an interpretation gap. “The color is darker” stands as an observation, while “the risk is higher” stands only under a color-to-risk mapping: explicit chains turn such omitted conditions into objects for verification.

3.2.2 Counter-Questions and Competing Explanations

The competing module poses three connected questions: whether the current observation suffices for the conclusion, which interpretation rules the inference depends on, and whether the same chart admits another evidence-based explanation under which the conclusion fails. From these questions the module generates a bounded candidate set

$$\Pi_t = \{\pi_0, \dots, \pi_K\}, \quad \pi_i = \langle O_i, B_i, p_i \rangle, \quad (3)$$

whose conclusions may support the current action, refute it, or support a different judgment; all candidates refer to the same complete chart. A competing chain is meaningful only if it differs substantially in observation correspondence, interpretation rule, applicability condition, or derivation scope: two chains that both accept “the curve rises” but adopt forward and reversed value mappings disagree precisely on the applicability of B . A reversed interpretation enters as a conditional hypothesis, unverified until the chart’s scales support it; the module may return insufficient support and need not produce an opposing conclusion for every task.

3.2.3 Normalization and Shared Rules

Generated chains are normalized over observation, rule, applicability scope, conclusion, and evidence dependence; paraphrases are merged, a rule used in several chains shares one verification state, and different entities or comparison relations keep separate evidence. Unlike self-consistency, which aggregates answers over reasoning paths (Wang et al., 2022), the object here is the rule each path depends on: repeated generations of the same rule add no evidential weight, and a new explanation matters only through different evidence or a new applicability problem. In deployment the candidate set is the frozen per-task pool of one to three O/B records pre-generated from the chart and public task (275 across the panel). So K is the pool size, with diversity a property of pool construction; the online arm re-derives chains at runtime under the same budget. This multi-candidate structure relates to multiagent debate (Du et al., 2023; Liang et al.,

2024) and to single-path self-refinement (Madaan et al., 2023), but the competitors disagree only over the interpretive bases of one rendered chart, and the adjudicator is an applicability test against that chart rather than a vote or LLM-judged winner.

3.3 In-Chart Applicability Verification

Verification first extracts the disputes that genuinely affect conclusions among the candidates, concentrating on the disputed mapping when chains agree on the observation, and on observation binding before rule applicability when the dispute concerns entity–value correspondence. For each chain the module outputs

$$V_i = (v_i^O, v_i^B, v_i^{\vec{B}}, E_i), \quad (4)$$

where each verification status takes the values support, refute, or undetermined, and E_i records the in-chart evidence. Observation O_i is checked for presence in the chart and for entity, value, time, and unit binding; rule B_i for support by the current chart and satisfaction of its applicability conditions; the derivation $O_i \wedge B_i \Rightarrow p_i$ for sufficiency of the confirmed premises, including candidate coverage, range, and task conditions. Verification always proceeds against the complete chart, and the evidence record contains the observation identifier, the content read, and the rule it supports or refutes. The deployed checklist operationalizes the three statuses: for O , located support or genuine contradiction, no inserted quantifiers or causal claims; for B , necessary-condition tests against public facts, including whether known labeled references would have the relations a proposed decoding predicts, and an explicit order comparison before any common scale is endorsed; for the derivation, coverage, range, and task-scope conditions as confirmed premises. Verification takes the normalized dispute propositions as input; the source role or repetition count of a candidate is never a credibility signal, and observations and interpretive conditions are confirmed before conclusions, so a preferred answer cannot justify a rule retroactively.

For rules incompatible over the same objects and scope, the method checks which in-chart information discriminates them: disputes with unresolved evidence stay open, and rules applying to different axes, series, or ranges have their applicability conditions split rather than forced to one choice. Verification may support a replacement rule or only negate the original—that bar height and visible

value ordering disagree does not entail “shorter means larger,” and the sound revision may be that bar height does not directly support quantity ordering. Rule revision therefore includes replacing a mapping and disqualifying an inference route.

3.4 Rule Revision and Persistent State Update

Each interpretation rule is maintained as a record

$$m_i = \langle \text{id}_i, B_i, \sigma_i, s_i, E_i, \Delta_i, v_i \rangle, \quad (5)$$

with id_i for later reference, B_i the interpretation rule, σ_i its applicability scope over chart objects, encoding components, and semantic conditions, $s_i \in \{\text{pending, active, disputed, revoked}\}$, E_i the supporting or refuting evidence, Δ_i linking original rule, revised rule, and change reason, and v_i the revision version. The state stores how the current chart is to be read and on what evidence; object choices, answers, and click actions are produced by task reasoning and recorded in the execution history. A rule’s scope binds by the conditions under which it holds—a position–value mapping to the numeric axis and series concerned, a color reading to the current legend and metric, an overall-trend rule to temporal order and comparison range; page changes unrelated to the task goal do not invalidate a rule.

Verification produces rule-revision proposals Δ_t^B , applied as $M_{t+1}^B = U(M_t^B, \Delta_t^B)$: a supported rule with an unevidenced proposal keeps the proposal pending; a refuted rule with a supported replacement activates the replacement in the affected scope; a refuted rule without a settled replacement is revoked and the interpretation gap recorded; effective conflicts enter disputed status while verification continues within budget; and rules applying to different objects or scopes are stored separately, not overwritten globally. The “active” status expresses the judgment under the evidence at hand; new effective evidence can change it, or show that an earlier verification was itself mistaken. State writes are issued only by the verification module; raw environment content never writes the state directly. After revocation, content, scope, and reason remain in the revision history; a reappearing rule must answer the recorded refutation or supply evidence that changes the applicability judgment—repetition, later position, or a more confident phrasing forms no new ground by itself. A revision record answers why the rule had to change, what interpretation is adopted now, and under which conditions it applies (Appendix A.2 illustrates).

An invalidated rule does not by itself condemn the current action, since $\neg B_0 \not\Rightarrow \neg C_q(a_0)$: after updating B , the method re-examines judgments that depended on it, retaining those other effective interpretations still support, so the admissible outcomes are a rule change with the action retained (a wrong reason for a right choice) or rule and action changing together; judgments already recorded are marked basis-changed for later re-confirmation, the interaction history stays intact, and the currently effective rules are organized by a separate read interface.

3.5 Reading and Using Rules over Continued Execution

Let the interpretation context c_t collect the relevant chart objects, encoding components, metrics, and task scope; the state reader returns $\mathcal{R}_t = \text{Read}(M_t^B, c_t)$, whose directly usable part is

$$\mathcal{R}_t^{\text{active}} = \{m_i \in M_t^B : s_i = \text{active} \wedge \text{Match}(\sigma_i, c_t) = \text{true}\}. \quad (6)$$

$\text{Match}(\sigma_i, c_t)$ evaluates the rule’s object, encoding component, metric, and time window against c_t and returns applicable, non-applicable, or undetermined; an undetermined match triggers verification rather than assuming a stored rule covers the new chart. Disputed rules never enter $\mathcal{R}_t^{\text{active}}$; they surface only as re-verification evidence in the read result, so interpretations still in force are separated from those that lost support, and revoked content never derives facts. When a chart has left the viewport, the reader keeps supplying applicable rules through the chart references established within the task, and new chart objects or encoding changes are bound and verified afresh.

The base agent produces its next proposal from the current observation, history, and rule view,

$$(\tilde{a}_t, \tilde{\pi}_t) = \mathcal{F}_\theta(q, o_t, H_t, \mathcal{R}_t), \quad (7)$$

where $\tilde{\pi}_t$ is a short interpretation record of the present judgment. Rules about the same chart can remain valid while the correct action changes with task conditions, so interpretation evidence is reused across steps while actions are re-derived from the current observation and the public task. A proposal depending on a revoked rule is returned the revision evidence and required to re-derive; a proposal bringing evidence sufficient to challenge an active rule re-enters competing verification, restraining old conventions from regaining force without grounds while preserving a channel for correcting

earlier mistakes. Ordinary actions follow the original interface; the persistent state acts through later contextual reads, with complete competing verification only when a trigger holds.

Three observable entry points start verification: the first chart-dependent judgment of a task, an interpretation conflict against the effective state or against in-chart evidence, and an applicability change of chart, encoding, series binding, or task scope. They depend on observations, task context, and action proposals; task correctness and hidden scoring never participate in triggering.

After a rule update, the agent re-judges the action and records *Keep* (the choice remains supported by an effective interpretation and the task conditions), *Revise* (relevant evidence negates the choice and an alternative gains the support the task requires), or *Unresolved* (a dispute that could still change the judgment persists, or necessary evidence is missing). Showing that a candidate exceeds the current choice refutes “the current choice is highest”; confirming the replacement still requires the public task’s comparison, a task-semantic check rather than one fixed formula. Unresolved status continues whole-chart verification within the remaining budget; an exhausted budget records the unfinished reason explicitly. Verification suggestions, actual revisions, and final submissions are recorded separately, and only executor operations constitute action outcomes.

3.6 Full Algorithm and Integration

Algorithm 1 (Appendix A) summarizes one CERS execution under budgets K and L for one verification node. The framework enters an existing agent at three points—the observation entrance establishes chart references, the pre-execution point performs verification and rule revision, and the later-reasoning entrance reads the applicable rules; planning, execution, and review modules share the rule state within one task, while action capabilities and the environment interface follow the original system. Explanation generation and verification can use different calls of the same frozen model, with deterministic code for record keeping, merges, version updates, scope retrieval, and explicitly computable checks; budgets and triggers are fixed before a run and charged to the method’s cost. The rule state is isolated per task: the runtime sees only the chart and interaction information the task permits; the paired chart, hidden answers, and scoring feedback stay on the evaluation side.

4 Experiments

This section reports the two evaluation deployments of CERS: a static paired panel that isolates the verification core, and a stateful online deployment that adds the persistent rule state inside a live browser harness.

4.1 Setting

The task basis is inherited from our companion benchmark MISVISAGENTBENCH, which pairs 140 executable web tasks—47 business, 35 environment, 19 health, 39 public-statistics, in 21 mechanism families—with a faithful chart and a misleading chart whose presentation violates an encoding convention while the data remain drawable.² Model inputs never contain gold labels, mechanism labels, the other chart variant, or scoring feedback.

Static panel. For every task the agent sees the benchmark’s misleading chart—scored against its frozen original label—plus the public goal and four options, and must select; no navigation occurs. Six configurations share identical inputs and decoding (temperature 0.7, top- p 0.8, top- k 20, fixed seed, thinking disabled): Direct and five READ–VERIFY–DECIDE configurations with frozen prompts, one verification round ($L = 1$), and three requests per task drawing on the frozen per-task pool of one to three O/B records pre-generated from the chart and public task (275 across the panel). V3 verifies O and B in distinct typed passes, V4 fixes explicit numeric references, V5 adds same-series query-point constraints with scale, time-window, and policy checks, V6 checks necessary conditions itemwise, and V7 keeps V4’s prompts with an abstention-aware decision repair phase. The five variants trade verification content against interface strictness: what is checked versus how answers are forced. The six configurations and prompts were frozen before the rerun and seed rounds, never modified after seeing results. Interface failures and empty selections remain in the 140 denominator; *robust accuracy* throughout is the fraction of tasks whose selection matches the frozen original label, a single full-denominator quantity. Frozen prompts, typed schemas, decision logs, the interface contract, and scoring code are in the supplementary material, so every number outside the three-seed replication can be recomputed from released artifacts once its license permits re-

²Companion submission under review; instance construction, taxonomy, and scoring are defined there.

	Original		27B rerun		Arm	Req. transfer (summed)				Selection acc.	Δ		
Config	Tar.	Δ	Tar.	Δ	p	95% CI	Direct (3 seeds)	Δ	p	3/task	95% CI	56.2 ± 1.1	
Direct	81	—	73	—	—	SC3 (majority)	65	—	—	0 extra	—	53.6%	—
V3	74	−7	75	+2	<.001	[+.09, +.22]	Tabular (bounded)	50	−15	.185/2/task	[−.02, +.13]	28.6%	—
V4	87	+6	86	+13	.0025	[+.05, +.23]	CERS V5 (3 seeds)	77	+12	.079/9/task	[−.01, +.18]	65.7 ± 0.7	+1
V5 (full)	96	+15	90	+17	.0013	[+.06, +.23]	Clean-chart control	75	+10	.044/1/task	[−.00, +.20]	82.1%	+
V6	84	+3	86	+13	<.001	[+.07, +.22]	Table 2	71	+6	.233	[−.04, +.16]		
V7	91	+10	89	+16	.0025	[+.05, +.23]	Matched-configuration replication (Appendix B): three request seeds for Direct and CERS, SC3	80	+15	.021	[−.02, +.21]		

Table 1: Static panel across three rounds (140 tasks; Qwen3.8-27B for the first two column groups, Qwen3-VL-8B-Instruct for the transfer group; temperature 0.7, top- p 0.8, top- k 20, fixed seed). “Target” counts selections matching the frozen original label; Δ is the paired net change against Direct within the same run; p and the exact 95% CI come from two-sided exact McNemar tests on wrong–target discordant pairs (McNemar, 1947), (empty selections and interface failures excluded from the test, retained in Δ). Rounds cost 8.86M/8.84M/8.14M tokens; error composition, stratification, and the development subset are in Appendix B.

lease.

Online stateful deployment. Twelve tasks spanning all four domains, in fixed identifier order, run as live browser sessions under a closed deployment: each task’s two chart variants execute under both ordinary and CERS arms, whose hooks, rule state, and re-derivation loops run inside the harness with the actor driving a (Playwright-controlled) browser. The submission interface is a strict typed protocol (typed_scalars_v2; at most eight actor calls, one re-verification round, temperature 0): a unit ends as a scored submission, an abstention with a reason, or a failure. Three actor systems are used—Qwen3.8-27B under strict response formats (Batch A), the same model behind a schema-guided proxy mirroring the sealed structural rules (Batch B), and a frontier model, gpt-5.6-terra, through an OpenAI-compatible gateway (Batch C, plus one stability rerun).

4.2 Static Panel Results

Table 1 reports the original round under Qwen3.8-27B: direct selection sits at 57.9% (81/140) and the full configuration V5 at 68.6% (96/140), a net gain of 15 decision points with 0 interface failures against Direct’s 1; its empty selections stay in the denominator, so the gain is not bought by abstaining. Because the panel is entirely single-decision-point, the gain cannot reflect information gained after selection: verification changes the decision

already available.

Table 2 reports a matched-configuration replication separating sampling variance from configuration effects: Direct and V5 are re-executed under three request seeds (12345, 22345, 32345) on a single-GPU deployment matching the frozen service except batch admission, so comparisons stay internal (Appendix B records the protocol). The same run supplies three baselines: an SC3 self-consistency vote over the Direct seeds (Wang et al., 2022) (majority among non-null selections, ties as abstentions), a tabular-answering baseline (Tonglet et al., 2026) (bounded transcription of comparison-relevant rows, then a decision from the table alone), and a clean-chart control on the paired clean charts.

The 27B columns of Table 1 report the single-run rerun that keeps model, seeds, and prompts frozen: direct selection drops from 81 to 73 of 140 under the identical configuration, quantifying agentic decoding’s sampling variance; comparisons must stay within-run. Within-run, V5 gains +17 paired decisions ($p = .0013$; 95% CI [+0.061, +0.231]), V4, V6, and V7 gain +13, +13, +16 ($p \leq 0.0025$ each), and V3 returns to +2 despite strongly directional wrong–target flips (30 against 6, $p < .001$; caption’s discordant-pair test) because its interface failures absorb the difference. The rerun’s paired transfers are one-directional (57 keep, 33 error-to-target, 14 target-to-error, 2 target-to-empty); interface failures concentrate in V3. No configuration can abstain (every unit must select), and the same directionality holds in every seed of the matched-

configuration replication, whose per-seed nets are reported in the baselines paragraph.

Baselines and controls. The clean-chart control quantifies misleading presentations’ execution-level damage: direct selection rises from 56.2% on misleading charts to 82.1% on the paired clean charts (paired net +33, $p < .001$, CI [+0.16, +0.27]). The two generic baselines do not close that gap: SC3 reaches 53.6%, indistinguishable from direct selection (paired net -3, $p = .375$)—consistent with Section 3.1, the object of correction is the interpretation rule, not the answer distribution; and the tabular-answering baseline abstains on 55 units with 18 transcription failures, selecting with 59.7% precision (40/67), a paired net of -3 ($p = .51$). Across the three seeds, direct selection lands at 56.2 ± 1.1 and CERS at 65.7 ± 0.7 points; the per-seed paired nets +12, +18, +15 pool to +45 decisions over 420 paired units (exact McNemar $p = 1.9 \times 10^{-5}$, 95% CI [+0.058, +0.150]).

Cross-model transfer. On Qwen3-VL-8B-Instruct, direct selection drops to 46.4% and the picture splits (Table 1, 8B columns): V7 gains +15 ($p = .021$; CI [+0.017, +0.210]), V5 gains +10 ($p = .044$), V4 and V6 remain within noise, and V3 collapses through 51 interface failures despite directional wrong-target flips (Appendix B.4). The panel’s exact tests are unadjusted for multiplicity; on the 8B columns we read V4–V7’s consistent directionality, not single-test significance. The separated verification prompts that produce directional flips on the 27B model overwhelm the 8B actor, so interface robustness is a first-class variable, and prompt content and interface contract must be evaluated jointly, not as one “method” factor.

4.3 Online Stateful Results

Because the deployment’s submission protocol is fail-closed, the primary online metric is the *wrong-submission rate*—the defense-side counterpart of agent abstention (Liu et al., 2026; Luo et al., 2026): across the four batches, ordinary execution produced six wrong submissions, all under the 27B actor on misleading charts, while CERS produced none in any batch or variant (0 of 96 defense-arm units); Table 3 gives the full four-batch decomposition. On clean charts the defense arm pays the mirror cost—6 and 5 correct submissions against the ordinary arm’s 10 (Batch C and its rerun)—the

price of fail-closed verification. Aggregate accuracy moves oppositely: ordinary execution loses 0–5 decision points on misleading charts against its own clean-chart performance, and the defense arm converts correct ordinary decisions into recorded abstentions rather than corrections. The paired effect is strictly one-directional in both frontier-model batches (10 and 13 preserved decisions become abstentions; the reverse never occurs; sign tests $p = .002$ and $p < .001$), with submission precision 100% (26/26 and 27/27). A pilot (6 tasks, gpt-5.6-terra) showed the same qualitative pattern (5 correct ordinary selections against 1 for CERS).

Interface decomposition. Structural and semantic failure modes separate by actor system: under the strict interface without guidance (Batch A) 20 of 24 defense-arm units die in engineering failures, while the schema-guided proxy (Batch B) zeroes the ordinary actor’s structural failures but cuts defense-arm engineering failures only modestly, and the direct gateway (Batch C) leaves honest abstention as the remaining mass (Appendix B.2).

A stability rerun of Batch C produces no wrong submission in either run: 37 of 48 units stay on the same side of the submission boundary, the rest move among abstentions, engineering failures, and call-limit terminations, and completion throughput varies (Appendix B.3).

5 Conclusion

Misleading charts are a runtime evidence problem: the agent’s own reading becomes part of its state, so a defense must compete explanations at the moment of reading and keep revised rules in force afterwards. CERS implements this pair through competing-explanation chains, in-chart verification, and a persistent rule state revised only through recorded Keep/Revise/Unresolved decisions. The static panel shows a robust-accuracy gain that replicates within-run, across seeds, and on an 8B agent; interface contracts co-determine outcomes as strongly as prompt content; and online, under a fail-closed protocol, the defense arm never submits a wrong answer, residual errors becoming legible abstentions. We release decision logs, rule states, and re-derivation traces so future work can measure which explanation an agent acted on and whether it survived the chart.

Limitations

This work studies defensive scaffolding around vision–language agents; its limitations follow from the same design choices that produce the results.

Scale and models. The static panel uses one 27B and one 8B open-weight VLM under fixed decoding; the online deployment adds one frontier model through a gateway. We do not claim the gains generalize to proprietary or future models, other agent frameworks, or tasks outside the four covered domains; the 8B transfer already shows that variant choice interacts with actor capability in ways we do not fully predict.

Baseline coverage. The panel compares against self-consistency, bounded tabular answering, and a clean-chart control; a prompt-instruction hardening control (instructing the actor to re-check axis labels) was not run and would test a complementary, model-side mitigation.

Measurement and variance. Robust accuracy is computed against the original option labels, and agreement checks verify that submitted options and rule states match the recorded logs; label agreement is not semantic verification, a run may rest on a misreading the frozen labels do not separate, recent benchmark-validity critiques (Lan et al., 2026) caution against reading any such score as grounded visual understanding, and the rule states we analyze are the agent’s own declared rules. The paired gains are within-run comparisons on a fixed task set, with sampling variance quantified three ways: the drop of direct selection from 81 to 73 of 140 between the two frozen rounds, the three-seed spread of ± 1.1 accuracy points around 56.2%, and McNemar tests with exact confidence intervals on paired decisions rather than cross-run rate differences; engine-level changes such as batch admission also shift individual-seed outcomes slightly, so cross-engine comparisons are avoided. The 24-task development subset was used to calibrate configuration emphasis and is scored separately throughout; still, all configurations were designed by the authors after inspecting aggregate failure patterns, so the panel should be read as a controlled comparison rather than a held-out evaluation.

Interface dependence and runtime. The online results are conditional on the strict submission protocol, whose semantic contract—exact public-leaf values, bounded summaries, count-coupled

responses—cannot be expressed in a response format; under earlier actor systems the same machinery died in engineering failures rather than abstentions, so the deployment demonstrates CERS under one operating point, not an interface-robust defense, and the negative environment-domain results and one-directional online paired effect are reported as found. The V3–V7 ladder varies verification content and interface prompts jointly, so the 8B attribution to interface robustness is argued from the failure-mode decomposition, not isolated by a prompt–interface cross ablation. Serving the six static configurations takes roughly $2.6\times$ as many requests as there are decisions (2,230 requests for 840 decisions), and the online arm operates within a fixed per-unit budget (one re-verification round, eight actor calls); agents under tighter latency or cost envelopes may not accommodate this overhead.

State integrity. The rule state is trusted harness memory: the isolation that keeps environment content from writing rules directly is assumed, and persistent-state poisoning attacks on agent memory (Louck, 2026)—and, more broadly, injection–execution dissociation (Leong, 2026)—fall outside the present threat model and remain targets for future work.

Ethics and reproducibility boundary. The defense direction of this work is protective, but the same observation—that faithful chart pixels admit misleading encodings that change executed actions—can inform adversarial chart design, so instances are released only through the benchmark’s own access process and no new manipulation techniques are published. The companion benchmark is under review and not yet public; until release the results cannot be independently recomputed from the logs alone, although the decision logs, prompts, and rule states of the original, replication, 8B, and online rounds are preserved verbatim in the supplementary material, and the three-seed replication survives as summary statistics and scoring outputs.

References

- Jason Alexander, Priyal Nanda, Kai-Cheng Yang, and Ali Sarvghad. 2024. [Can gpt-4 models detect misleading visualizations?](#) In *2024 IEEE Visualization and Visual Analytics (VIS)*.
- Shubham Bharti, Shiyun Cheng, Jihyun Rho, Jianrui Zhang, Mu Cai, Yong Jae Lee, Martina Rau, and Xi-

- aojin Zhu. 2024. [Chartom: A visual theory-of-mind benchmark for llms on misleading charts](#). *Preprint*, arXiv:2408.14419.
- Graziano Blasilli and Marco Angelini. 2026. [True \(vis\) lies: Analyzing how generative ai recognizes intentionality, rhetoric, and misleadingness in visualization lies](#). In *arXiv (Cornell University)*. Cornell University.
- Sizhe Chen, Julien Piet, Chawin Sitawarin, and David Wagner. 2024a. [Struq: Defending against prompt injection with structured queries](#). *Preprint*, arXiv:2402.06363.
- Sizhe Chen, Arman Zharmagambetov, Saeed Mahloujifar, Kamalika Chaudhuri, David Wagner, and Chuan Guo. 2024b. [Secalign: Defending against prompt injection with preference optimization](#).
- Zixin Chen, Sicheng Song, Kashun Shum, Yanna Lin, Rui Sheng, Weiqi Wang, and Huamin Qu. 2025. [Unmasking deceptive visuals: Benchmarking multimodal large language models on misleading chart question answering](#).
- Amit Kumar Das, Mohammad Tarun, and Klaus Mueller. 2025. [Charts-of-thought: Enhancing llm visualization literacy through structured data extraction](#). In *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, *PrePrints* 5555, pp. 1-11, 2025.
- Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2023. [Improving factuality and reasoning in language models through multiagent debate](#). arXiv:2305.14325. *Preprint*, arXiv:2305.14325.
- Ivan Evtimov, Arman Zharmagambetov, Aaron Grattafiori, Chuan Guo, and Kamalika Chaudhuri. 2025. [Wasp: Benchmarking web agent security against prompt injection attacks](#). *Preprint*, arXiv:2504.18575.
- Xueyu Hu, Tao Xiong, Biao Yi, Zishu Wei, Ruixuan Xiao, Yurun Chen, Jiasheng Ye, Meiling Tao, Xiangxin Zhou, Ziyu Zhao, Yuhuai Li, Shengze Xu, Shenchi Wang, Xinchun Xu, Shuofei Qiao, Zhaokai Wang, Kun Kuang, Tieyong Zeng, Liang Wang, and 10 others. 2025. [Os agents: A survey on mllm-based agents for computer, phone and browser use](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7436–7465, Vienna, Austria. Association for Computational Linguistics.
- Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. 2023. [Large language models cannot self-correct reasoning yet](#). *Preprint*, arXiv:2310.01798.
- Sam Johnson, Viet Pham, and Thai Le. 2025. [Manipulating llm web agents with indirect prompt injection attack via html accessibility tree](#).
- Rachneet Kaur, Nishan Srishankar, Zhen Zeng, and Sumitra Ganesh. 2026. [Chartagent: A multimodal agent for visually grounded reasoning in complex chart question answering](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 18488–18555, San Diego, California, United States. Association for Computational Linguistics.
- Jing Yu Koh, Robert Lo, Lawrence Jang, Vikram Duvvur, Ming Lim, Po-Yu Huang, Graham Neubig, Shuyan Zhou, Russ Salakhutdinov, and Daniel Fried. 2024. [Visualwebarena: Evaluating multimodal agents on realistic visual web tasks](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 881–905, Bangkok, Thailand. Association for Computational Linguistics.
- Zixuan Lan, Luzhe Sun, Matthew R. Walter, and Jiawei Zhou. 2026. [Seeing without looking: Do vision-language benchmarks really test vision?](#) arXiv:2605.22903. *Preprint*, arXiv:2605.22903.
- Jun Wen Leong. 2026. [Injection-execution dissociation: A mechanistic evaluation of persistent memory attacks and defenses in stateful llm agents](#). arXiv:2605.08442. *Preprint*, arXiv:2605.08442.
- Yin Li. 2025. [Decomposing llm self-correction: The accuracy-correction paradox and error depth hypothesis](#). arXiv:2601.00828. *Preprint*, arXiv:2601.00828.
- Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Shuming Shi, and Zhaopeng Tu. 2024. [Encouraging divergent thinking in large language models through multi-agent debate](#). pages 17889–17904.
- Maxim Lisnic, Cole Polychronis, Alexander Lex, and Marina Kogan. 2023. [Misleading beyond visual tricks: How people actually lie with charts](#). pages 1–21.
- Xun Liu, Yi Evie Zhang, Vira Kasprova, Parisa Rabhani, Pardis Sadat Zahraei, Tianyu Zhang, Ali Ebrahimpour-Boroojeny, and Varun Chandrasekaran. 2026. [Agentabstain: Do llm agents know when not to act?](#) arXiv:2607.10059. *Preprint*, arXiv:2607.10059.
- Yedidel Louck. 2026. [Securing llm-agent long-term memory against poisoning: Non-malleable, origin-bound authority with machine-checked guarantees](#). arXiv:2606.24322. *Preprint*, arXiv:2606.24322.
- Han Luo, Bingbing Wen, and Lucy Lu Wang. 2026. [Agentic abstention: Do agents know when to stop instead of act?](#) arXiv:2606.28733. *Preprint*, arXiv:2606.28733.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder,

- Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. [Self-refine: Iterative refinement with self-feedback](#). arXiv:2303.17651. *Preprint*, arXiv:2303.17651.
- Ridwan Mahbub, Mohammed Saidul Islam, Md Tahmid Rahman Laskar, Mizanur Rahman, Mir Tafseer Nayeem, and Enamul Hoque. 2025. [The perils of chart deception: How misleading visualizations affect vision-language models](#). In *2025 IEEE Visualization and Visual Analytics (VIS)*.
- Ridwan Mahbub, Mohammed Saidul Islam, Md Tahmid Rahman Laskar, Mizanur Rahman, Mir Tafseer Nayeem, and Enamul Hoque. 2026. [Chart deception in vision-language models: From vulnerability to mitigation](#). *Preprint*, arXiv:2607.22600.
- Quinn McNemar. 1947. [Note on the sampling error of the difference between correlated proportions or percentages](#). In *Psychometrika*.
- Khandaker Abrar Nadib, Marina Kogan, Alexander Lex, and Maxim Lisnic. 2026. [Guardrail selection in line charts to contextualize persuasive visualizations](#). In *Computer Graphics Forum (EuroVis)*, 45(3), 2026.
- Swaroop Panda. 2026. [Making ai agents evaluate misleading charts without nudging](#).
- Mizanur Rahman, Huan Wu, Arash Asgari, Enamul Hoque Prince, and Laleh Seyyed-Kalantari. 2026. [Same chart, different story: Bias in vision-language chart interpretation](#). arXiv:2609.26210. *Preprint*, arXiv:2609.26210.
- Christian Seto, Jacqueline Nguyen, Jiayi Hong, and Ross Maciejewski. 2026. [Llms have visualization literacy: Now what? experiments exploring llm visualization evaluation capabilities](#). In *arXiv (Cornell University)*. Cornell University.
- Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. [Reflexion: Language agents with verbal reinforcement learning](#). *Preprint*, arXiv:2303.11366.
- Jonathan Tonglet, Tinne Tuytelaars, Marie-Francine Moens, and Iryna Gurevych. 2026. [Protecting multimodal large language models against misleading visualizations](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8329–8349, San Diego, California, United States. Association for Computational Linguistics.
- Jonathan Tonglet, Jan Zimny, Tinne Tuytelaars, and Iryna Gurevych. 2025. [Is this chart lying to me? automating the detection of misleading visualizations](#). arXiv:2508.21675. *Preprint*, arXiv:2508.21675.
- Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yuan Yang, Jingsen Zhang, Zhiyuan Chen, Jikai Tang, Xu Chen, Yankai Lin, Wayne Xin Zhao, Zhewei Wei, and Ji-Rong Wen. 2024. [A survey on large language model based autonomous agents](#). *Frontiers of Computer Science*, 18(6).
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2022. [Self-consistency improves chain of thought reasoning in language models](#).
- Zhun Wang, Vincent Siu, Zhe Ye, Tianneng Shi, Yuzhou Nie, Xuandong Zhao, Chenguang Wang, Wenbo Guo, and Dawn Song. 2025. [Agentvigil: Automatic black-box red-teaming for indirect prompt injection against llm agents](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 23159–23172, Suzhou, China. Association for Computational Linguistics.
- Chen Henry Wu, Rishi Shah, Jing Yu Koh, Ruslan Salakhutdinov, Daniel Fried, and Aditi Raghunathan. 2024. [Dissecting adversarial robustness of multimodal llm agents](#). *Preprint*, arXiv:2406.12814.
- Qiusi Zhan, Zhixiang Liang, Zifan Ying, and Daniel Kang. 2024. [Injecagent: Benchmarking indirect prompt injections in tool-integrated large language model agents](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 10471–10506, Bangkok, Thailand. Association for Computational Linguistics.
- Yanjie Zhang, Yafei Li, Rui Sheng, Zixin Chen, Yanna Lin, Huamin Qu, Lei Chen, and Yushi Sun. 2026. [Navigating the mirage: A dual-path agentic framework for robust misleading chart question answering](#). In *arXiv (Cornell University)*. Cornell University.
- Wei Zhao, Zhe Li, Peixin Zhang, and Jun Sun. 2026. [Clawguard: A runtime security framework for tool-augmented llm agents against indirect prompt injection](#). *Preprint*, arXiv:2604.11790.
- Chenyu Zhou, Huacan Chai, Wenteng Chen, Zihan Guo, Rong Shan, Yuanyi Song, Tianyi Xu, Yingxuan Yang, Aofan Yu, Weiming Zhang, Congming Zheng, Jiachen Zhu, Zeyu Zheng, Zhuosheng Zhang, Xingyu Lou, Changwang Zhang, Zhihui Fu, Jun Wang, Weiwen Liu, and 2 others. 2026a. [Externalization in llm agents: A unified review of memory, skills, protocols and harness engineering](#). arXiv:2604.08224. *Preprint*, arXiv:2604.08224.
- Wei Zhou, Xuanhe Zhou, Shaokun Han, Hongming Xu, Guoliang Li, Zhiyu Li, Feiyu Xiong, and Fan Wu. 2026b. [Are we ready for an agent-native memory system?](#) arXiv:2606.24775. *Preprint*, arXiv:2606.24775.

A Execution Loop and Worked Example

Algorithm 1 summarizes one execution of CERS under the candidate budget K and the review budget L for a single verification node.

A.1 Mechanistic Details

Rule merging by object and scope. “Darker means higher risk” and “a deeper color stands for a more severe risk” map to one rule when object and

scope coincide, whereas comparing Illinois with Kansas and Illinois with Delaware under the same legend are two comparisons that shared rules do not merge.

Update objects and chain exclusion. When a chain is challenged, the method distinguishes three objects of update: an *observation error* (a misread number, wrong entity binding, or wrong range) revises the affected O and re-verifies dependent interpretations, never charging unreliable observations to B ; an *inapplicable rule*—whose observation stands while its observation-to-semantics mapping disagrees with the current chart—is revoked or scope-narrowed; and *insufficient derivation* leaves both supported while the task conclusion remains unproven, so the module adds the missing comparison or keeps an undetermined status, as when a higher candidate refutes “the current candidate is highest” without confirming the alternative is globally highest. Conflicting conclusions between candidate chains establish only that a conflict exists; a chain is excluded only when chart-supported evidence negates its observation, its rule applicability, or its derivation conditions.

Revision record example. For a reversed linear axis, the revision record replaces “higher position, larger value” because the visible scale increases downward, states that an upward move of the bound series now reads as a decrease, and scopes this to the current chart, axis, and series.

A.2 A Worked Example of Revision and Cross-Step Reuse

We illustrate the mechanism on the chart semantics of task pub013 from our execution data, which asks the agent to choose the highest-risk record among Illinois, Kansas, and Delaware; the case material records that the map legend assigns lighter yellow to higher risk and deep red to lower risk. The walkthrough below demonstrates the method and does not add experimental results. Suppose the current proposal selects Kansas under

$$\pi_0 : \underbrace{\text{Kansas is darker}}_{O_0} \wedge \underbrace{\text{darker means higher risk}}_{B_0} \Rightarrow C_q \quad (8)$$

The competing module asks why a darker color should mean a higher risk; the model generates legend-dependent interpretations on the same complete map and compares the legend with the default color convention. If verification confirms that the

chart does not support B_0 , the rule loses applicability on this map and the state records the original interpretation, the revision reason that the convention contradicts the current legend definition, the current interpretation that candidate colors are read through the verified legend mapping, the scope of the current map and risk metric, and the condition for further updates, namely a different encoding or an effective refutation of the legend reading. The agent then re-compares the candidates with B_1 and the observed colors and completes the judgment the public task requires; the concrete option is an output of this reasoning, while the rule state keeps the interpretation basis of the current map. When the agent enters later pages or re-visits the chart-dependent choice, what it reads is that rule and its revision reason: a new explanation that merely repeats “deep red usually means more danger” must answer the recorded legend evidence and cannot restore B_0 directly, while a re-reading that finds the earlier legend verification mistaken is allowed to revise again. The process records three distinct things: what changed in the interpretation rules, whether the current action must change, and which rule version later steps actually used.

B Additional Results and Boundary Statements

B.1 Where the Gains Concentrate

Stratified by mechanism family, V5’s net gains concentrate where misleading power comes from scope and reference manipulation: the subset-extrapolation family (22 tasks) yields +18 and +11 across the two rounds, while non-linear-axis and dual-axis families are neutral to −6. By domain, business carries the largest share (+16 and +4), public statistics is neutral then positive (0 and +9), health is +6 in both, and the environment domain is the only negative one in both rounds (−5, −9). The development subset contributes at most two decision points to each headline gain, and costs remain bounded: 8.84M and 8.14M tokens across the rerun and 8B rounds (2,230 and 2,184 requests, against KS’s nominal full-run 2,240; shortfalls reflect early-terminating failures) against 8.86M in the original round. The requests serve 840 decisions each (six configurations $\times$ 140 tasks), a $2.6\times$ average request overhead over one-request direct selection, with frozen prompts and no parameter updates. The pool records were pre-generated before evaluation from each chart and its public task—136 tasks by

an earlier frontier-model batch and 4 by Qwen—state observations and interpretation rules rather than conclusions, are not a strictly blind construction (a minority of records carry task-directional hints that verification must re-justify against the chart), and their one-off generation cost is excluded from the per-task request counts.

B.2 Interface Decomposition Detail

The three actor systems separate structural from semantic failure modes. Under the strict interface without guidance (Batch A), 20 of 24 defense-arm units die in engineering failures (17 of them schema rejections at the defense stage), and the ordinary arm ends in two call-limit exhaustions and three wrong submissions on misleading charts. A schema-guided proxy mirroring the sealed structural rules (Batch B) drives the ordinary actor’s structural failures to zero (12/12 on clean charts) but cuts defense-arm engineering failures only modestly (21 of 24): the binding constraint is semantic, since exact public-leaf values, bounded summaries, and the question–response count coupling cannot be expressed by any response format—components added for self-correction become the exposed surface, as [Wu et al. \(2024\)](#) analyze for compound systems. Running the frontier model directly (Batch C) removes the structural interface almost entirely (engineering failures 4 of 24), leaving honest abstention—12 `unresolved_public_evidence` verdicts, mostly on misleading charts, where verification cannot close the gap between the printed chart and the required scope within the re-verification budget.

B.3 Stability Rerun

Re-running Batch C under the identical configuration leaves every submission correct: 37 of 48 units stay on the same side of the submission boundary (31/48 under the four-way classification), and the differences move units among abstentions, engineering failures, call-limit terminations, and submissions—never into a wrong submission, so the “never submit a wrong answer” property is stable while completion throughput is not. Token budgets are comparable across the two runs (1.80M input and 129K output tokens, 322 attempts and 242 browser operations, versus 1.81M and 115K, 331 and 254), confirming that the stateful machinery operates within a fixed per-unit allowance comparable to the static panel’s per-task cost.

B.4 Cross-Model Transfer Analysis

On the 8B open-weight VLM, direct selection drops to 46.4% and the picture splits (Table 1, 8B columns): V7 gains +15 ($p = .021$; CI [+0.017, +0.210]), V5 gains +10 ($p = .044$), V4 and V6 remain within noise, and V3 collapses—51 of 140 runs end in interface failures of the schema-typed protocol, against 8 and 6 in the other rounds, for a net of −15 despite directional wrong–target flips. The interaction is instructive: the separated verification prompts that produce directional flips on the 27B model overwhelm the 8B actor, so interface robustness is a first-class variable and prompt content and interface contract must be evaluated jointly rather than as one “method” factor.

B.5 Online Batch Decomposition

B.6 Matched-Configuration Seed Replication

The revision experiments rerun direct selection and the full configuration under three request seeds (12345, 22345, 32345) on a single-GPU deployment whose engine configuration matches the frozen service except for batch admission (`max_num_seqs=8` instead of 1), so all comparisons reported for these runs are internal to that configuration. The three plain runs also provide the three samples for the self-consistency baseline at no additional requests, with majority vote among non-null selections and ties recorded as abstentions. The tabular-answering baseline re-requests each task in two stages under the same engine: a transcription request that renders, from the image, a compact markdown table limited to the series and x positions the public task’s comparison involves (at most 24 rows), and a decision request answered from the table alone, so the decision stage never sees the image. The clean-chart control runs plain direct selection on the paired clean charts and is scored against the original labels, the protocol established by the companion benchmark’s online evaluation.

B.7 Development Subset and Stratified Results

The 24-task development subset is scored separately in every round. In the 27B rerun the full configuration V5 gains +17 paired decisions, +16 on the remaining 116 tasks and +1 on the development subset; the original round splits as +13 and +2, and the 8B round as +9 and +1. The environment domain is the only one in which V5 loses

Unit class	Cor.	Wrong	Abst.	Bug.	Fail.
Batch A (Qwen3.8-27B, strict)				13/21/4	5/5/3
Ordinary × misleading	7	3	–	–	–
Ordinary × clean	10	–	–	–	–
CERS × misleading	0	–	1	–	–
CERS × clean	3	–	–	–	–
Batch B (Qwen3.8-27B, schema-guided proxy)					
Ordinary × misleading	7	3	–	–	–
Ordinary × clean	12	–	–	–	–
CERS × misleading	1	–	–	–	–
CERS × clean	2	–	–	–	–
Batch C (gpt-5.6-terra, formal run)					
Ordinary × misleading	8	–	3	–	–
Ordinary × clean	10	–	2	–	–
CERS × misleading	2	–	7	–	–
CERS × clean	6	–	4	–	–
Batch C rerun (stability)					
Ordinary × misleading	10	–	1	–	–
Ordinary × clean	10	–	2	–	–
CERS × misleading	2	–	5	–	–
CERS × clean	5	–	4	–	–

Table 3: Online stateful deployment: 12 tasks spanning four domains, each run on the misleading chart and its clean pair under ordinary execution and the CERS arm (48 units per batch). “Cor.” counts submitted actions matching the original label; “Wrong” counts submitted actions that do not (all six occur in the ordinary arm on misleading charts in Batches A and B); “Abst.” recorded abstentions; “Bug.” re-verification budget exhaustion; “Fail.” engineering and response-format failures (including actor-call-limit terminations). All submissions in Batch C and its rerun are correct (26/26 and 27/27).

against direct selection in both rounds compared here (−5 in the original round, −9 in the 8B round). At the family level, subset-extrapolation tasks contribute +18 (original round) and +11 (8B round) for V5; dual-axis and non-linear-axis families are mildly negative for V5 in the second model round (−5 and −6), and bar-height tasks are near-neutral (−3 and −2).

B.8 Non-Scoring Outcomes

Interface failures are runs in which the agent never produces a schema-valid selection. Their counts by round (original / rerun / 8B) are Direct 1/0/1, V3 6/8/51, V4 0/2/2, V5 0/3/4, V6 2/2/8, and V7 0/1/2, none discarded from the denominator. Empty selections (completed runs with no recorded option) by round (original / rerun / 8B) are Direct 0/0/3, V3

13/21/4, V4 9/11/6, V5 6/5/8, V6 13/12/5, and V7 5/5/3, likewise retained in the denominator. V3’s interface failures are not discarded: they remain in the denominator, which is why its large error-to-target correction counts on the 27B model (30 forward against 6 reverse, exact McNemar $p = .0001$) reduce to a net +2. The 8B actor’s failure mode is dominated by responses that violate the typed-record contract of the verification pass rather than by timeouts or truncations.

B.9 Online Deployment Details

Each online unit executes one task on one chart variant under one system inside a sealed environment: public inputs are mounted read-only, rule hooks intercept chart evidence entering the context, and the unit ends in exactly one of the recorded terminal statuses (submitted, abstention with reason, re-verification budget exhausted, or engineering failure). The strict interface (`typed_scalars.v2`) permits at most eight actor calls and one re-verification round per unit at temperature zero. The three actor systems differ only below the interface: a local 27B model under strict response formats (Batch A), the same model behind a schema-guided proxy mirroring the sealed structural rules (Batch B), and the frontier model reached directly through the gateway (Batch C and its stability rerun). Across the two frontier-model batches the defense arm read its persistent rule state 48 and 46 times across its 24 units, never cooccurred with a wrong submission, and every per-unit persistence-causality flag reads `not_established`; the ordinary arm produced exactly six wrong submissions, all chart-induced, all under the 27B actor on misleading charts, and none under the frontier actor. On clean charts the defense arm submits 3, 2, 6, and 5 correct actions in Batches A, B, C, and the rerun, against the ordinary arm’s 10, 12, 10, and 10—the mirror of the misleading-chart conversions.

B.10 Boundary Statements

Three boundary statements delimit the claims. First, the per-unit causal persistence effect is recorded as *not established* in every online unit: the harness shows that revised rules were re-read and that outcomes changed between arms, but a single-unit causal attribution of outcome change to persistence is not identified by this design. Second, the stability rerun’s unit-level agreement (37/48) bounds the reliability of any single-batch online estimate; disaggregated four-way cells shift by at

most three units, and no flip ever crosses into a wrong submission. Third, the online deployment’s paired effect is one-directional: the defense arm converts ordinary decisions into abstentions (10 and 13 per batch) and never the reverse, so the deployment should be read as a safe-fail characterization of the defense under the strict protocol, not as an accuracy improvement result.

Algorithm 1 CERS execution loop

- 1: inputs: task q ; base agent $\mathcal{F}_\theta$ and the original action interface; chain budget K ; review budget L .
 - 2: initialize rule state $M \leftarrow \emptyset$; execution history $H \leftarrow \emptyset$.
 - 3: **for** each step t **do**
 - 4: receive observation o_t ; update chart references and interpretation context c_t .
 - 5: read applicable rules, revision evidence, and disputes from M .
 - 6: $(\tilde{a}_t, \tilde{\pi}_t) \leftarrow \mathcal{F}_\theta(q, o_t, H, \mathcal{R}_t)$.
 - 7: **if** a trigger holds: first chart-dependent judgment, interpretation conflict, or applicability change **then**
 - 8: expand the current explanation into $O \wedge B \Rightarrow C_q(\tilde{a}_t)$.
 - 9: generate at most K substantially different competing chains on the same complete chart.
 - 10: normalize candidates; merge repeated chains and shared rules.
 - 11: verify observations, rule applicability, and derivation sufficiency.
 - 12: apply adoption, revocation, replacement, or scope adjustment by the evidence.
 - 13: re-derive the current action from the updated state; within L , resolve remaining key disputes.
 - 14: **end if**
 - 15: **if** the proposal reuses a revoked rule without new evidence, return the revision record and require re-derivation; with an effective new challenge, re-verify the rule.
 - 16: with KEEP or REVISE, execute the approved action through the original interface; with exhausted budget, record UNRESOLVED and stop the decision.
 - 17: append the true observation, proposal, rule versions, action, and feedback to H .
 - 18: **end for**
 - 19: save the task’s rule revisions and execution trajectory; close the task’s rule scope.
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